DATA BASES – PROF. D. M. BRAGA – A. A. 2021/22

CALL 3 – JUNE 13TH 2022 – DURATION: 1h 30m

SQL (24 points)



The schema below describes the database of the Visa office in the Indian Consulate of Milan. Applicants request visas and, after some investigation, either a Visa is issued, associated to the request, or the request is rejected. The values of Outcome are either "*processing*" (for requests under evaluation), "*granted*", or "*rejected*". Visa types can be *Single entry*, *Business*, *Tourist*, *Residence*, *Short Stay*, …. ; please note that it is possible that applicants request a Visa of some type (say, *Residence*) and they are granted one of a "weaker" type (say, *Short Stay*).

APPLICANT (PassportNum, Name, BirthDate, Email, Nationality)
REQUEST (ReqCode, PassportNum, RequestDate, RequestedType, ContactInIndia, AdditionalNotes, Outcome)
VISA (Number, ReqCode, GrantedType, ValidityStartDate, ValidityEndDate)

1. Briefly but precisely explain what is extracted by the following view and query [**5 p**]:

a.
```
create view acc( typ, num ) as
    select GrantedType, count(*)
    from REQUEST natural join VISA
    where Outcome = "granted"
        and GrantedType = RequestedType
    group by GrantedType
```

b.
```
select PassportNum, Name
from APPLICANT A natural join REQUEST
group by PassportNum
having count( * ) >= 5
    and count( distinct RequestDate.day ) = 1
    and count( distinct RequestDate.month ) = 1
```

2. Extract in SQL:

- **a**. The *oldest* Italian applicant at the time of the request (*assume for simplicity that a subtraction between two dates returns an interval expressed as the number of days in between*); [**5 p**]
- **b**. The Passport number and Name of the applicants who submitted 4 or more requests and were never granted a Visa of the type that they requested; [**5 p**]
- **c**. The *acceptance rate* of each Visa type (i.e., the ratio of the *granted visas* to the *requests* for the type). [**5 p**]

3. Assume that the view defined in **1.a** is materialized. Formulate a set of triggers that incrementally maintain the materialization of the view in response to *insertions* on table VISA. [**4 p**]

Conceptual Design (8 points)

4. Design an E-R model for the database described in the following scenario:

A tailor-made clothing shop for men and women has a website with a catalog of models, that are starting points for orders of suits and dresses. Models have a code, name, description, type (suit, shirt, pants, outerwear,…), and if they are for males/females. Every model can be produced in different fabrics and colors, and every available fabric has a code, a name, a description, and a color.

Clients can order custom-made items of clothing of a specific model, and every ordered item has a progressive number (unique for each given model), the client's measurements (sleeve length, chest around, waist, hips, thigh, …), the chosen cloth and color, the price, and possibly extra annotations. For clients, we store phone number, email, and the latest shipping address and measurements. Orders are either completed or in preparation; orders in preparation have an expected time of delivery, completed orders have the date when the good was shipped, the carrier name and their tracking ID.

- *The project is evaluated for completeness, correctness, legibility, minimality and self-explanatory nature.*
- *Remember to specify the **identifiers** of all entities and the **cardinalities** of all associations.*